

to 2000 Hz. The resonance frequency is determined by mounting the transducer at the centre of the PCB. At the resonant frequency, the PCB is continuously vibrated for 30 minutes, after which the critical bare board test and micro-section is performed to observe any deformation in the barrels.

Humidity storage tests. This is to determine the effect of storage in moisture, which could result in a reduction in the performance of the PCB.

Other tests that are performed are out-gassing, dielectric withstanding voltage tests, insulation resistance tests, interconnection resistance tests, hot oil dip tests, peel strength tests, network simulation, bond strength tests, bare board tests and visual inspection.

An overview of the test pattern used for achieving high-reliability PCBs at ISRO

ISRO has an in-house PCB manufacturing plant that is called a 'protoshop.' For testing, 11 samples of a PCB are submitted to the quality assurance division, where engineers conduct Group-1 to Group-7 tests on them. These tests reveal the quality of the processes employed in fabricating the PCBs. Also, the tests check whether the process identification documentation and flow-chart has been made in a proper manner, whether non-conformance management has been recorded properly, and whether feedback from various in-line quality control (QC) at each stage of manufacturing has been recorded. After all the tests, the manufacturer of the PCB gets the information from ISRO's QC.

In Group-1, pre-conditioning, cleaning and visual-eye examination are performed. The 11 samples or boards given for testing should have been manufactured in different batches. Out of 11, ten PCBs are subjected to destructive testing and one PCB is kept as a reference to note down the deviations after each test that the board undergoes.

In Group-2, the thermal cycling test

Critical aspects to consider while designing and manufacturing PCBs used in spacecraft

- Long life
- Consistent performance
- High reliability and excellent workmanship
- Ability to withstand exposure to environmental conditions



A space vehicle at the time of launch (Courtesy: www.isro.org)

is done, where the PCB is subjected to +125 to -55 degrees centigrade for hundred cycles. Similarly, after the thermal cycling is done, the test engineers carry out visual examination, bow and twist to check for continuity, inner layer IR checks, high voltage tests, peel strength, re-work simulation, and plating thickness, which is done by a micro-section. After each test that is carried out, the monitoring test is performed to check if anything has happened to the PCB during these tests.

In Group-3, the current carrying capacity for each trace design is fixed – this is one of the critical aspects to be measured, since the trace or copper track must not fail during operation. The board is also subjected to long-term damp heat to simulate various humidity conditions.

In Group-4, hot storage and cold storage tests are carried out. In the hot storage test, the PCB is subjected to a temperature of 85 degrees centigrade for 14 days, whereas in the cold storage test, it is kept for 24 hours at -40 degrees centigrade. A vibration

test is also carried out to detect plated-through holes, especially in multi-layer boards.

In Group-5, high-temperature dip tests are done. Generally,

when any re-work is done on PCBs, they are subjected to thermal excursion. Soldering and desoldering of a component causes thermal excursion. This has to be verified by carrying out high-temperature dip tests. Generally, in any of the highly-reliable PCBs, the solder flow test is performed only once at 288 degrees centigrade for ten seconds, but for space-grade PCBs, the same test is done six times, i.e., the PCBs used for space applications are dipped in solder six times to see if plated-through holes, pads and other things are intact. This can be verified by performing the bow and twist test, peel strength test and micro-section of the PCB.

In Group-6, dimensional verification and solderability tests are carried out – for both hole and pads surface. These tests check if solder masks have been properly covered, because outerspace is highly contaminated with gases. Out-gassing of PCB materials is also checked.

Finally, Group-7 consists of the final tests that are carried out on PCBs. These tests check whether the components have been assembled properly on the PCB or not.

'Reliability' is the key word

In the world of defence and aerospace, reliability is very important. The fact that the systems used here are of national importance, and that they also have the potential to cause destruction to human life or property, is what makes safety such a high priority. ●

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